UCLA Math 131A, Winter 2025
Midterm 2 Practice

Instructor: Linfeng Li

Friday, February 21

Please print your name and student ID below.

Name: ______________________________

Student ID: ______________________________

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- Calculators, notes, and textbooks are **not** allowed.
- Good luck!

Question	Points	Score
1	10	
2	7	
3	9	
4	7	
5	7	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement.

   (i) True False A sequence in $\mathbb{R}$ is Cauchy if and only if it is convergent.

   (ii) True False If a sequence is unbounded, then it diverges to ∞ or diverges to $-\infty$.

   (iii) True False If a sequence (s_n) is bounded, then $\limsup s_n$ and $\liminf s_n$ are real numbers.

   (iv) True False If $(s_n)_{n\in\mathbb{N}}$ is convergent, then every subsequence $(s_{n_k})_{k\in\mathbb{N}}$ is convergent.

   (v) True False A sequence $(s_n)_{n\in\mathbb{N}}$ converges to 1 if and only if $(|s_n|)_{n\in\mathbb{N}}$ converges to 1.

2. (7 points) Prove that the sequence $((-1)^n)_{n\in\mathbb{N}}$ is not Cauchy by the definition of a Cauchy sequence.

3. Let $(s_n)_{n\in\mathbb{N}}$ be defined recursively by $s_1 = 1$ and

$$s_{n+1} = \frac{3s_n + 2}{s_n + 2}, \quad \text{for all } n \in \mathbb{N}.$$

   (a) (3 points) State the Monotone Convergence Theorem.

   (b) (3 points) Use induction to prove that $0 \le s_n \le 2$, for all $n \in \mathbb{N}$.

   (c) (3 points) Prove that $(s_n)_{n\in\mathbb{N}}$ is convergent.

4. (7 points) Let $(s_n)_{n\in\mathbb{N}}$ be a sequence of integers. Prove that if the sequence $(s_n)_{n\in\mathbb{N}}$ is Cauchy, then the sequence is eventually constant, i.e., there exists some $N \in \mathbb{N}$ such that $s_n = s_m$ for all $n, m \geq N$.

5. (7 points) Let $(s_n)_{n\in\mathbb{N}}$ be a sequence with $\lim_{n\to\infty} s_n = 2025$. Prove that there exists a subsequence $(s_{n_k})_{k\in\mathbb{N}}$ such that $s_{n_k} > 2025 - \frac{1}{k}$ for all $k \in \mathbb{N}$.